Angular Acceleration: Examples

SYNOPSIS

A practice sheet, not new material. These are the two angular-acceleration problems from the middle of **Note 2**, pulled out and worked longhand so there is a clean copy of each to follow.

Both are rotational kinematics with $\alpha \neq 0$, which means the three equations from Note 1's table — the ones that sat there looking like the linear ones — are finally the equations you need. The only real work is at the two ends: converting rev/min into rad/s at the start, and remembering at the finish which of $a_t = r\alpha$ and $a_c = r\omega^2$ the question actually asked for. They are different accelerations, at right angles to each other, and on the ultracentrifuge they differ by a factor of a million and a half.

BEFORE YOU START — PREDICT

A spinning rotor is speeding up. A point on its rim therefore has two accelerations at once. Write one sentence naming them and saying which way each one points.

PERFORMANCE EXPECTATIONS

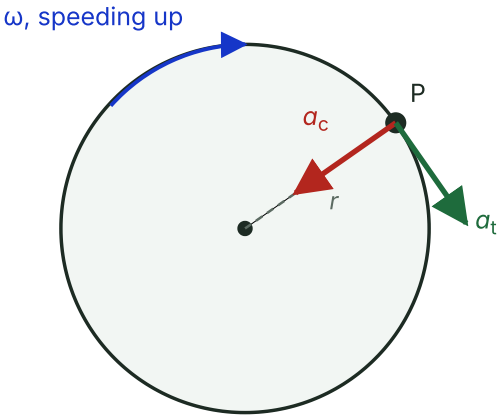
Convert rev/min to rad/s. Apply the rotational kinematics equations when the angular acceleration is not zero. Distinguish tangential from centripetal acceleration for a point on a rotating body, and compute each. Express a centripetal acceleration as a multiple of g.

WATCH · THE SHEET WORKED THROUGH

Recording to come. Print the sheet (link at the bottom of this page), make your prediction, then work the items as the video goes through them.

Two accelerations, at right angles

Before either problem, be clear which acceleration is which. A point on a rotating body that is *speeding up* has both — one along the direction of travel, one straight at the axis — and they are completely different sizes.



$a_t = r\alpha$ along the motion; zero if $\alpha = 0$

$a_c = r\omega^2$ at the axis; never zero while turning

One has α in it and one has ω^2. On the ultracentrifuge below they come out as 3.14 and 4 930 000, in the same units.

The same point, the same instant, two accelerations. a_t says the speed is changing; a_c says the *direction* is. Reading one question's answer off the other formula is the single most common way to lose this problem.

1. The angular velocity of a rotating rigid body increases from 450 to 1400 rev/min in 135 s. (A) What is the angular acceleration of the body, in rad/s²? (B) Through what angle, in rad, does it turn in this 135 s?
2. An ultracentrifuge accelerates from rest to 100 000 rpm in 2.50 min. (A) What is the average angular acceleration, in rad/s²? (B) What is the tangential acceleration, in m/s², of a point 4.50 cm from the axis of rotation? (C) What is the centripetal acceleration of that point at full rpm, in m/s² and in multiples of g? (D) What total distance, in metres, does that point travel?

Show the working

Looking ahead

Both of these appear in **Note 2**, in the section on rotational motion with angular acceleration; this sheet is the clean longhand copy. If you want the derivations behind them rather than the arithmetic, that is Note 2. **Note 3** leaves kinematics and asks what it takes to *produce* an angular acceleration, which is where the moment of inertia comes in.

BACK TO YOUR PREDICTION

The two are the **tangential acceleration**, which points along the direction of travel and exists only because the rotor is speeding up, and the **centripetal acceleration**, which points straight at the axis and is there whenever the thing is turning at all. They are at right angles, so neither one adds to or cancels the other — and in question 2 they come out as 3.14 m/s² and 4.93×10^6 m/s². If you named only one, it was almost certainly the centripetal one.

Check yourself

1. Convert 1400 rev/min to rad/s.

_____ rad/s

2. A wheel turning at a steady rate, with no angular acceleration, has

- ○ no acceleration at any point on it
- ○ a centripetal acceleration but no tangential acceleration
- ○ a tangential acceleration but no centripetal acceleration
- ○ both, and they are equal

3. A body starts at 47.1 rad/s and accelerates at 0.737 rad/s². To find the angle turned in 135 s you must

- ○ use $\frac{1}{2}\alpha t^2$, since the angle starts at zero
- ○ use $\omega_0 t + \frac{1}{2}\alpha t^2$, since it is already turning at the start
- ○ use ωt with the final ω
- ○ use αt^2

4. A point 4.50 cm from the axis turns at 10 472 rad/s. What is its centripetal acceleration, in multiples of *g*? Give your answer in units of 10^5 *g*.

_____ $\times 10^5\ g$

Check Clear

Your notes

IN CLASS

The sheet — blank (PDF)

print or open on a tablet; make your prediction, then work the items.

AFTER THE MODULE

Completed sheet (PDF)

released when Module 5 closes.

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